Aalto University
School of Science

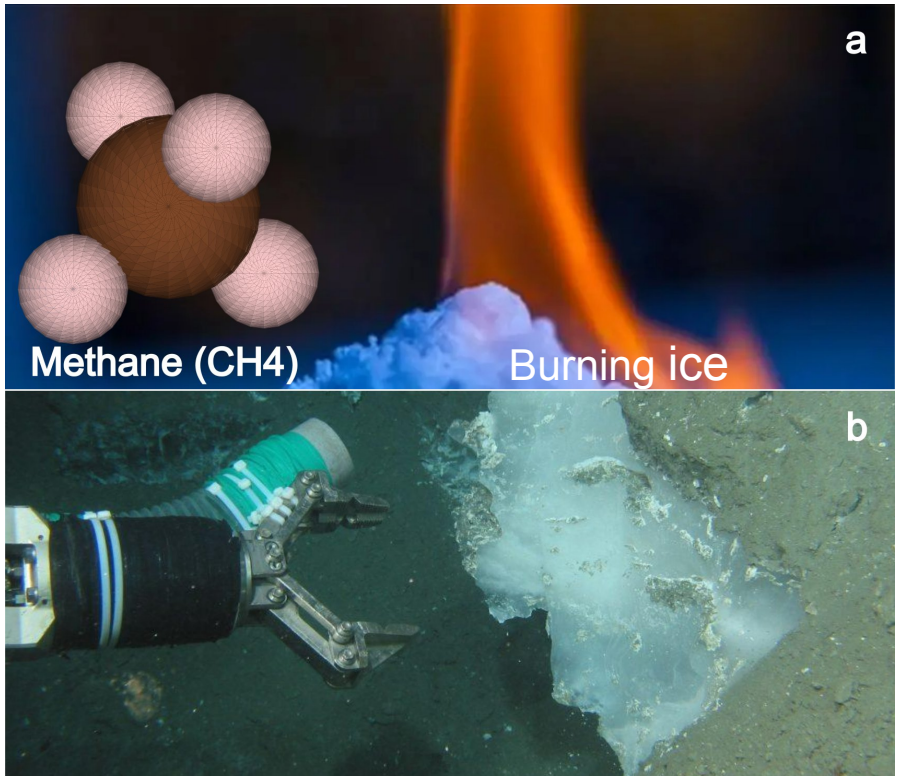
High-Resolution AFM and Simulation of THF Clathrate Hydrate Growth

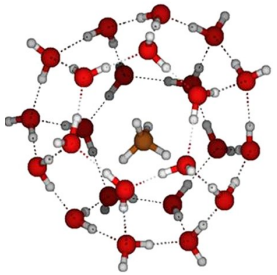
Jie Huang
Surfaces and Interfaces at the Nanoscale (SIN), Aalto University, Finland

ACSIN-16, Himeji

What are clathrate hydrates?

Clathrate Hydrates



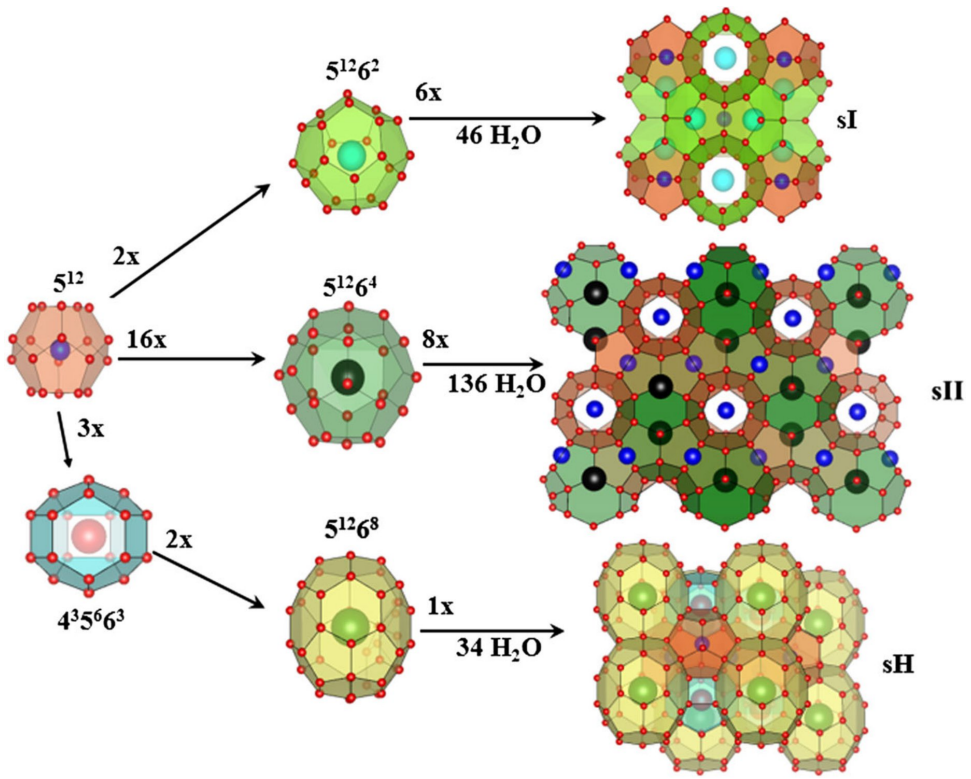


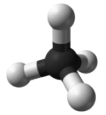
- Water molecules form cage-like structure, which **traps guest molecules** like methane.
- Methane hydrates can be found under the seafloor (**low temperature** below 10°C and **high pressure** above 2 MPa)
- The potential to become a **future energy resource**.

A!

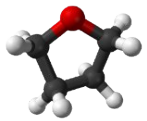
[1] J. Phys. Chem. A. 2012;116(29):7742-7745. doi:10.1021/jp304229p, [2] Alamy, via BBC News 2017; and [3] JAPAN Forward, 2021.

Structures of clathrate hydrates

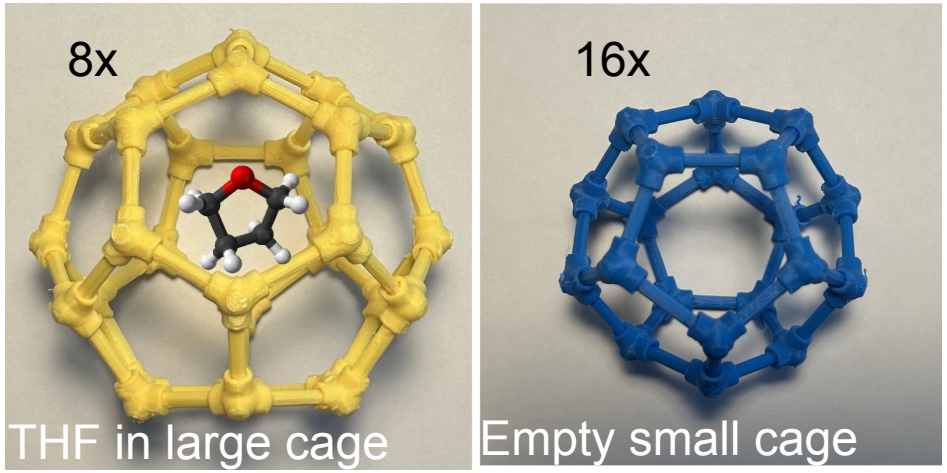




Methane hydrates: structure **sI**.



THF (C_4H_8O) hydrates: structure **sII**.



vertex: O in water molecules.
edge: Hydrogen bond between water molecules.
The H atoms are omitted.

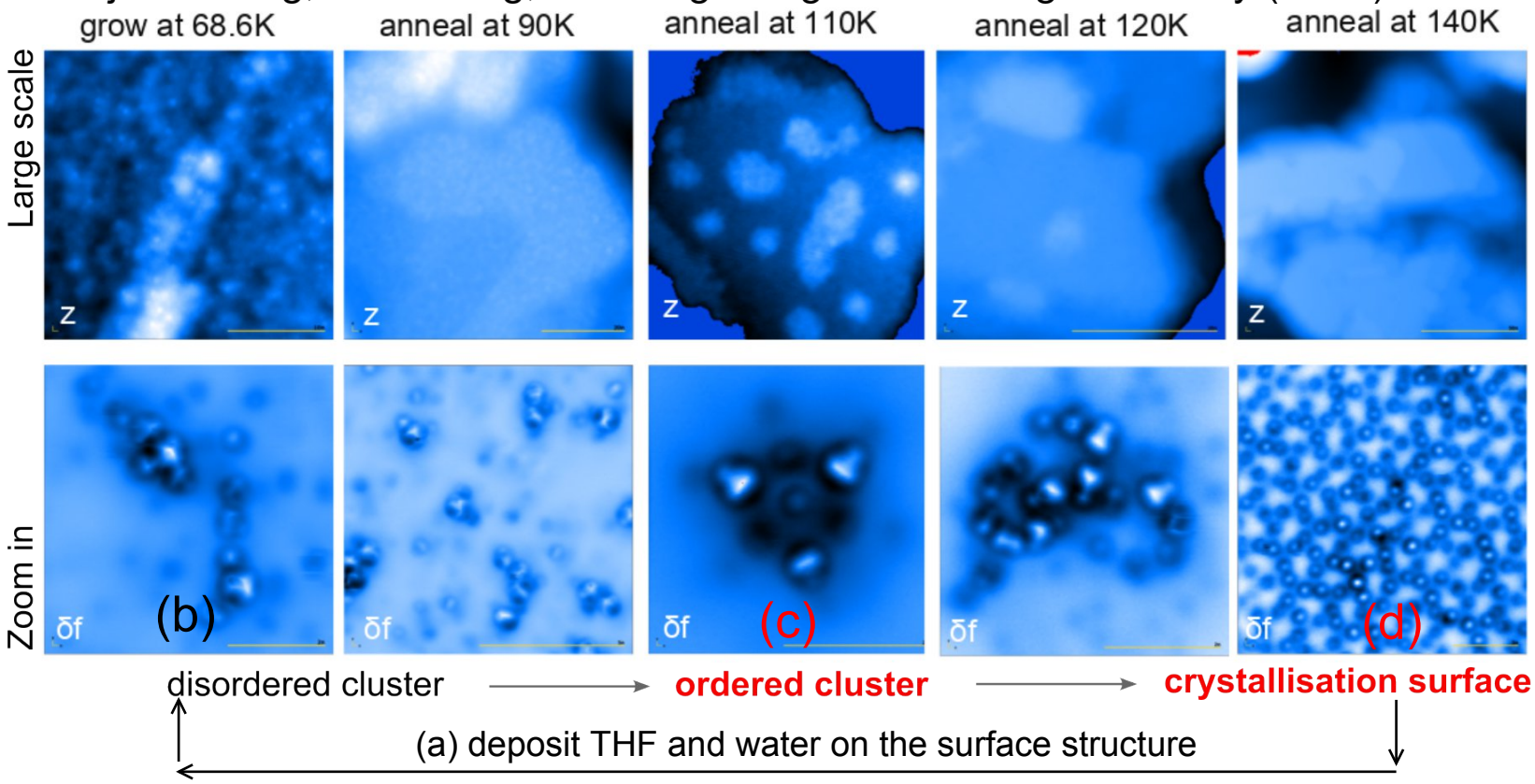
The formation of **THF hydrates does not require high pressure** (atmospheric pressure, bellow 4 °C).

A!

Can we see THF Clathrate Hydrates Growth Through AFM?

THF hydrate surface at different temperatures

Yuejian Zhang, Jiani Hong, and Ying Jiang from Peking University (2025)



Questions:

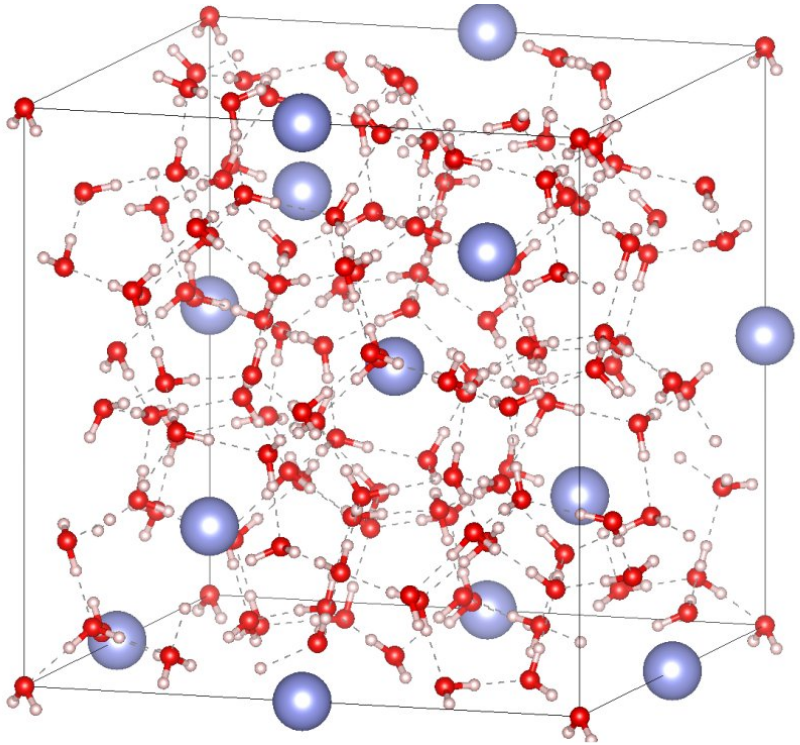
Can we understand each stage in this process?

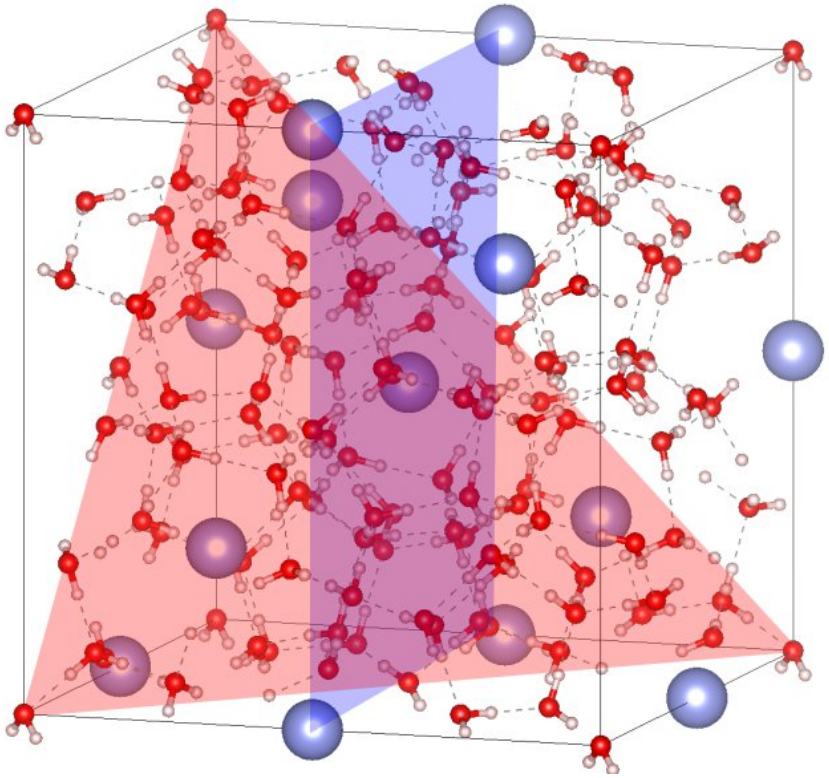
What are the atomic structures behind these stages?

Depositing THF and water on the surface structure (d), annealing at different temperatures, **disordered clusters (b)** become **ordered clusters (c)**, and back to a stable surface forms (d).

AFM Simulations for Structure Discovery

Starting point of simulation : the bulk system



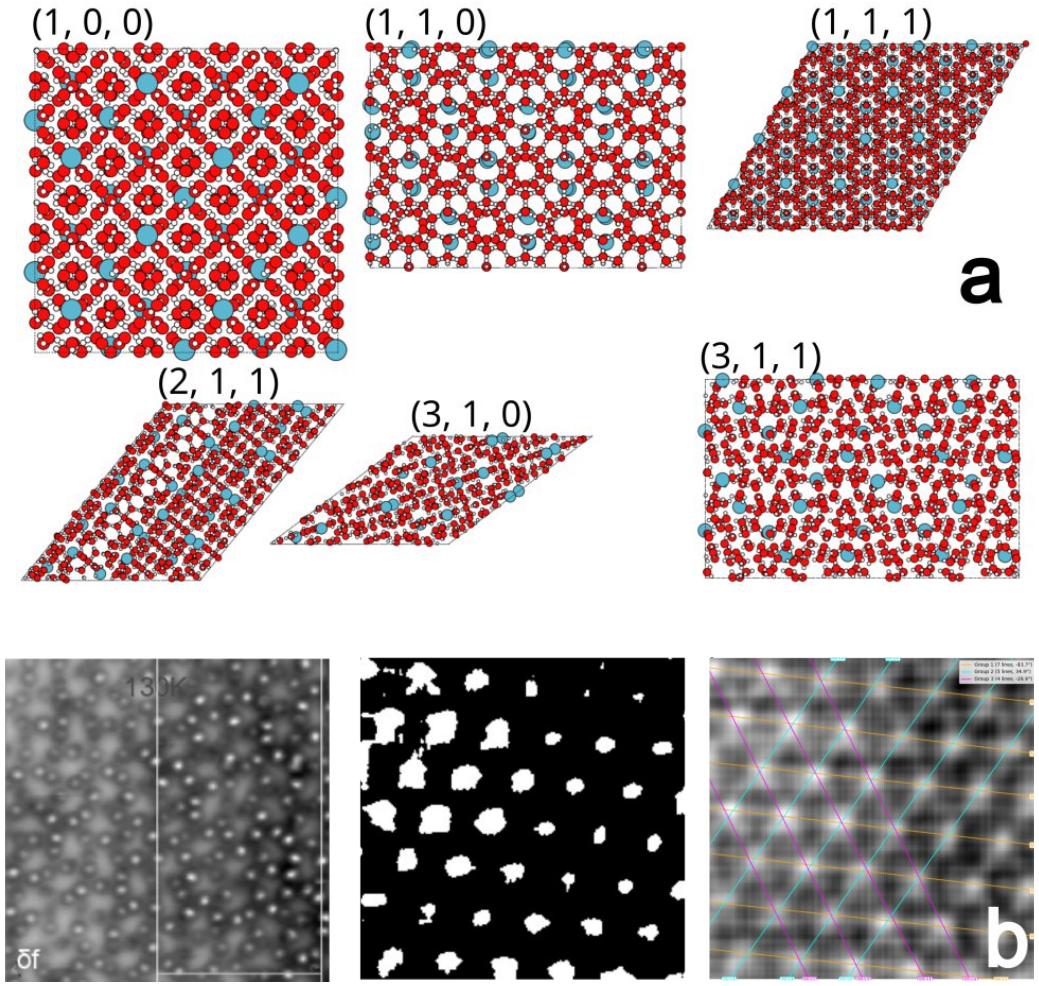


Cutting along different planes gives different surface slabs.
What is the orientation of the experimental surface?

A!

[1] J. Chem. Phys. 138, 124504 (2013)

From Bulk to Surface Structure: Orientation Determination



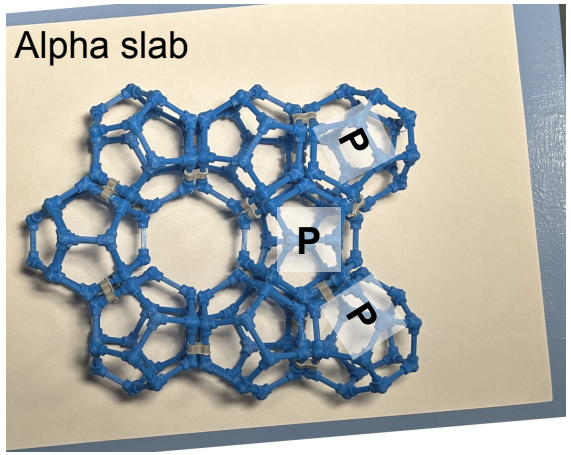
Analyzing the AFM image of surface structure and compare feature spacing and symmetry with experiment.

The orientation of experimental surface is (111).

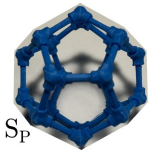
Simulation for (1,1,1)	AFM experiment for surface structure:
Group1: 11.84 ± 0.19 Angstrom	Group1: 11.94 ± 0.59 Angstrom
Group2: 11.74 ± 0.18 Angstrom	Group2: 11.32 ± 1.31 Angstrom
Group3: 11.80 ± 0.13 Angstrom	Group3: 11.84 ± 0.57 Angstrom
Angle(1,2) = 59.13°	Angle(1,2) = 61.39°
Angle(1,3) = 62.00°	Angle(1,3) = 56.84°
Angle(2,3) = 58.87°	Angle(2,3) = 61.77°

A!

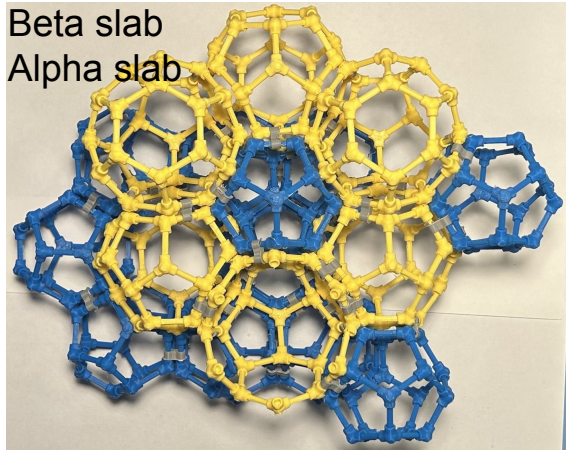
Substructures in (111) orientated surfaces



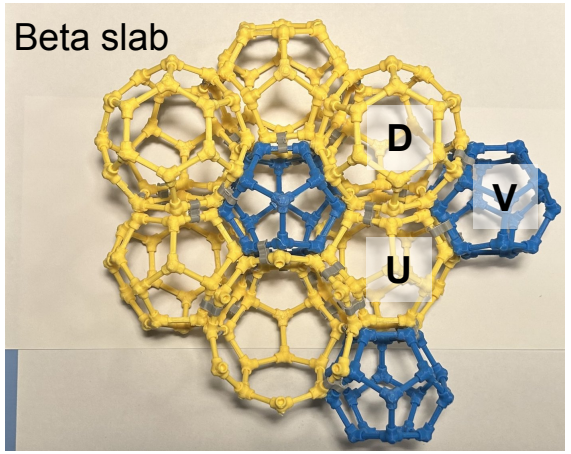
Alpha slab contains small cages, which **share the same orientation P** on the normal direction.

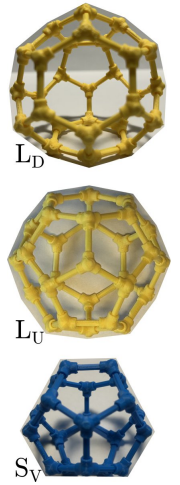


P



Two slabs can be **stacked** and then **repeated**.





D

U

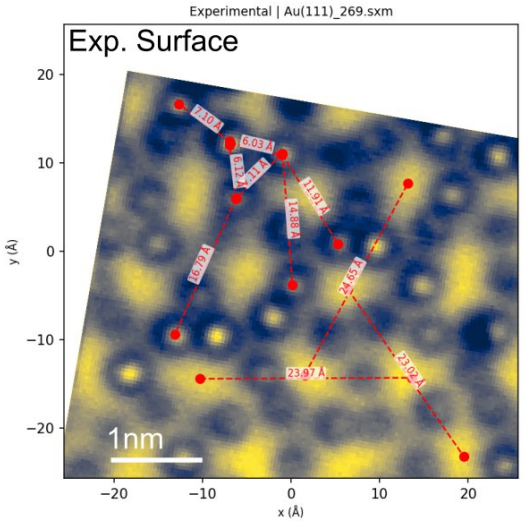
V

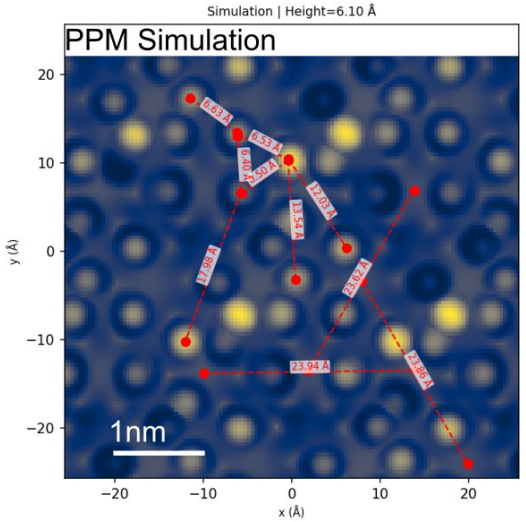
Beta slab contains small and large cages.
Large cages have two orientations, **D** and **U**
Small cages share another orientation **V**.

A! **We group the structure in (111) orientation into two slabs and four kinds of cages.**

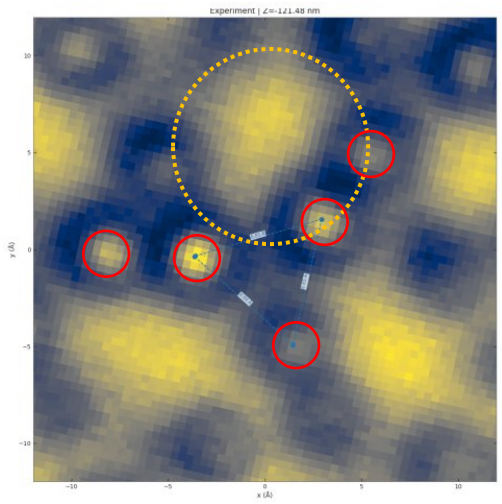
The surface structure candidates

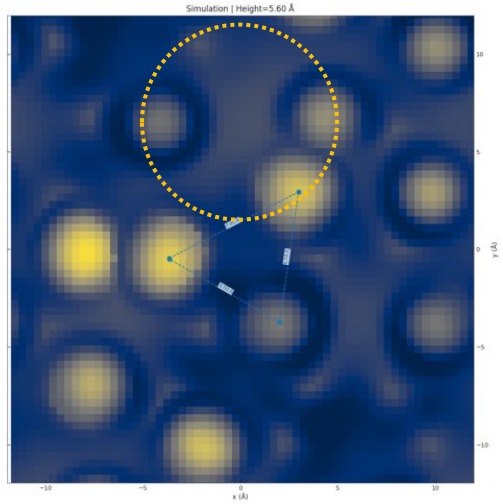
Simulation for the surface structure

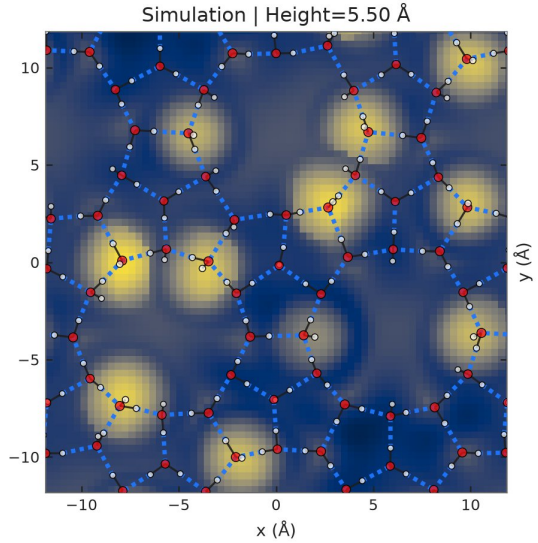


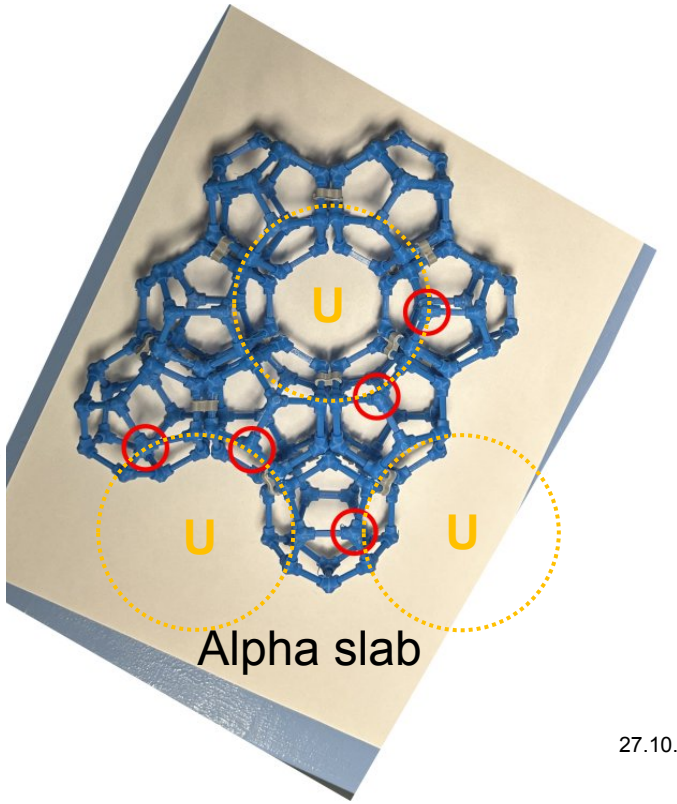


1. The **overall distances for similar features match.**.
2. Round features correspond to **upward free OH**.
3. The background **bright shadows** correspond the **open large cages U**.



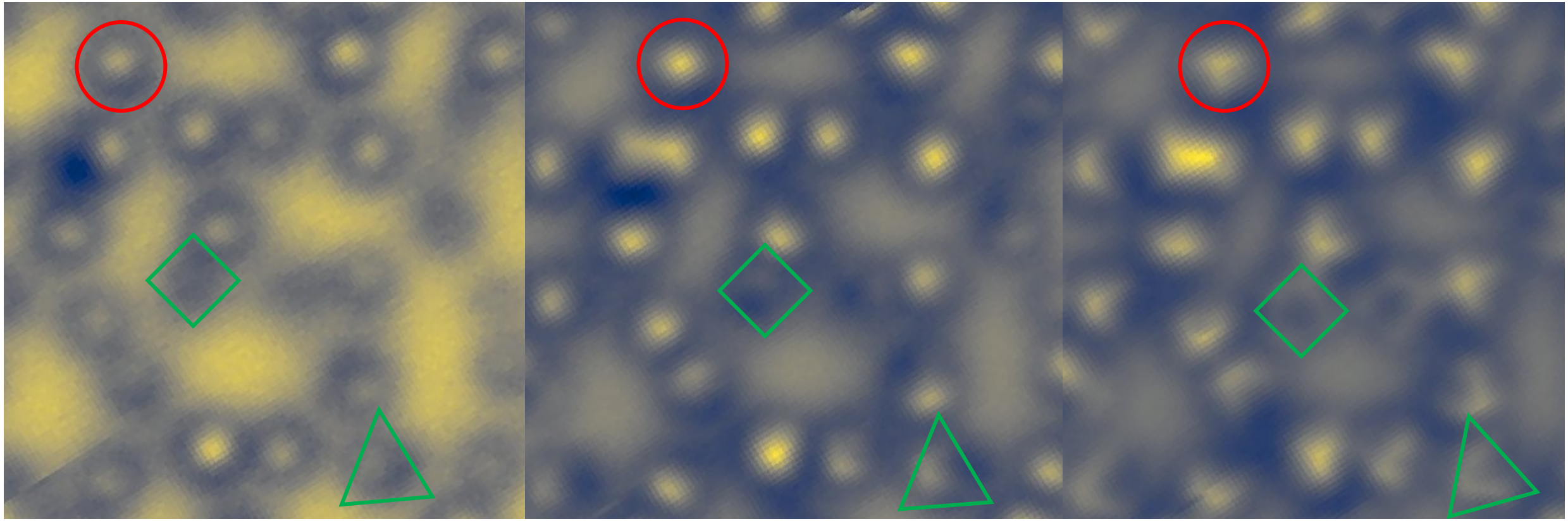






A!

Other features on the surface structure

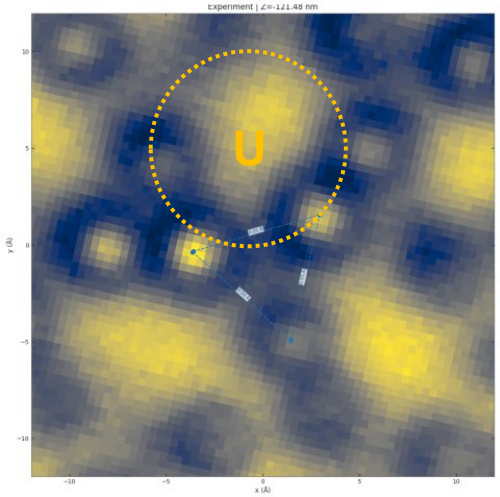


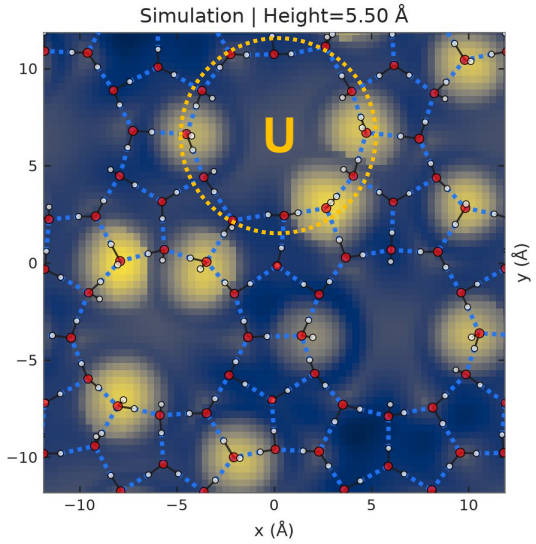
Apart from the feature of upward free OH, there are also some other features look like square and array.

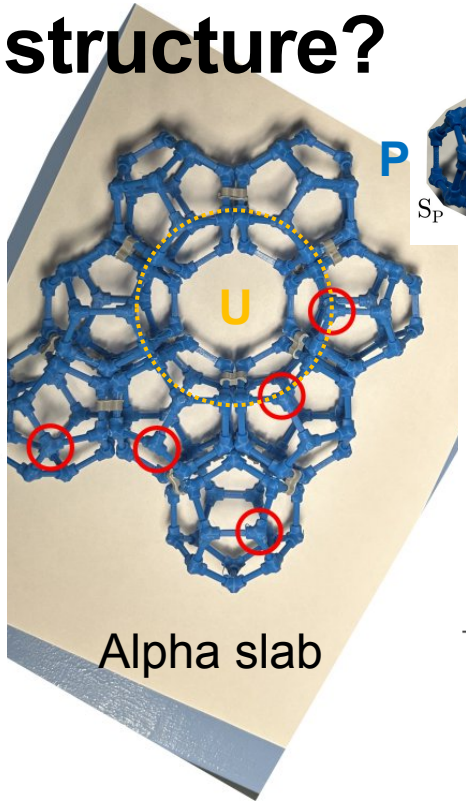
These are the feature from THF molecules in different rotations.

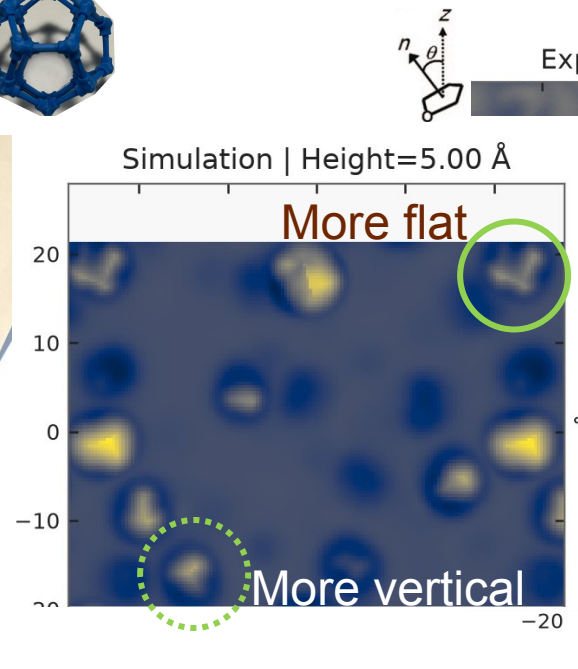
A!

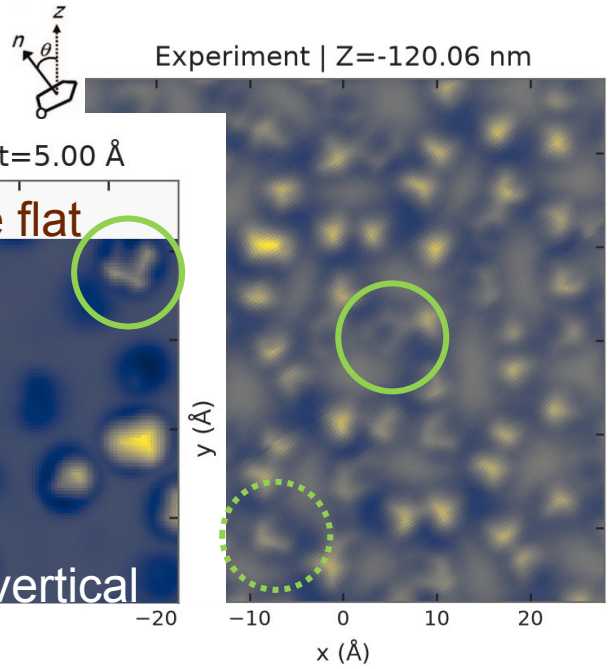
What we learn from surface structure?











1. Round features are attributed to water with **upward free OH** on the top of **P-orientated small cages** in **Alpha slab**
2. The bright shadow feature corresponds to the open **U cages**.
3. Extra THF are also found in some regions of surface area.

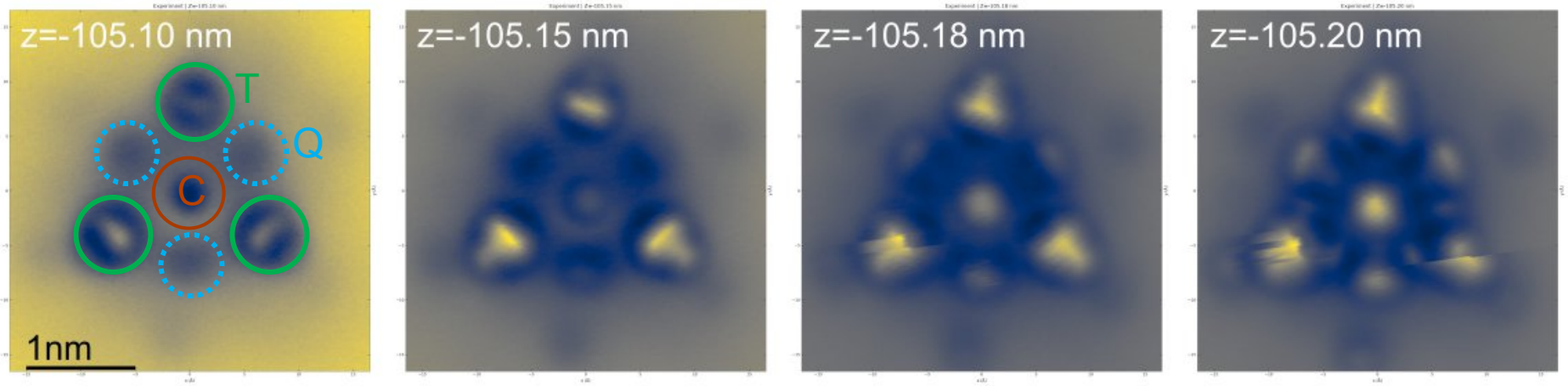
The experimental surface is a complete Alpha slab with additional THF in open large cages.

A!

The cluster structure candidates

The Cluster Structure

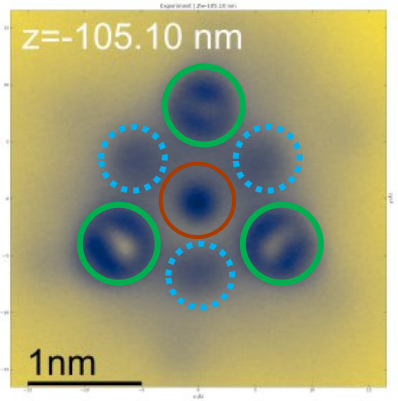
Cluster



What we know from experimental images?
These features are **three folds symmetric**.
The center feature (C) keeps round at different heights.
The outer most feature (T) shows line-to-T height response.
The middle features (Q) also show similar line-to-T features after.

Hypothesis:
1. C corresponds to free OH in a water molecule.
2. T features correspond to THF molecules.
3. Q features also correspond to THF molecules at lower positions.

Initial guess for the cluster

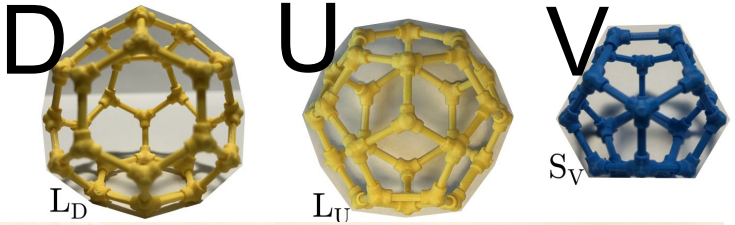


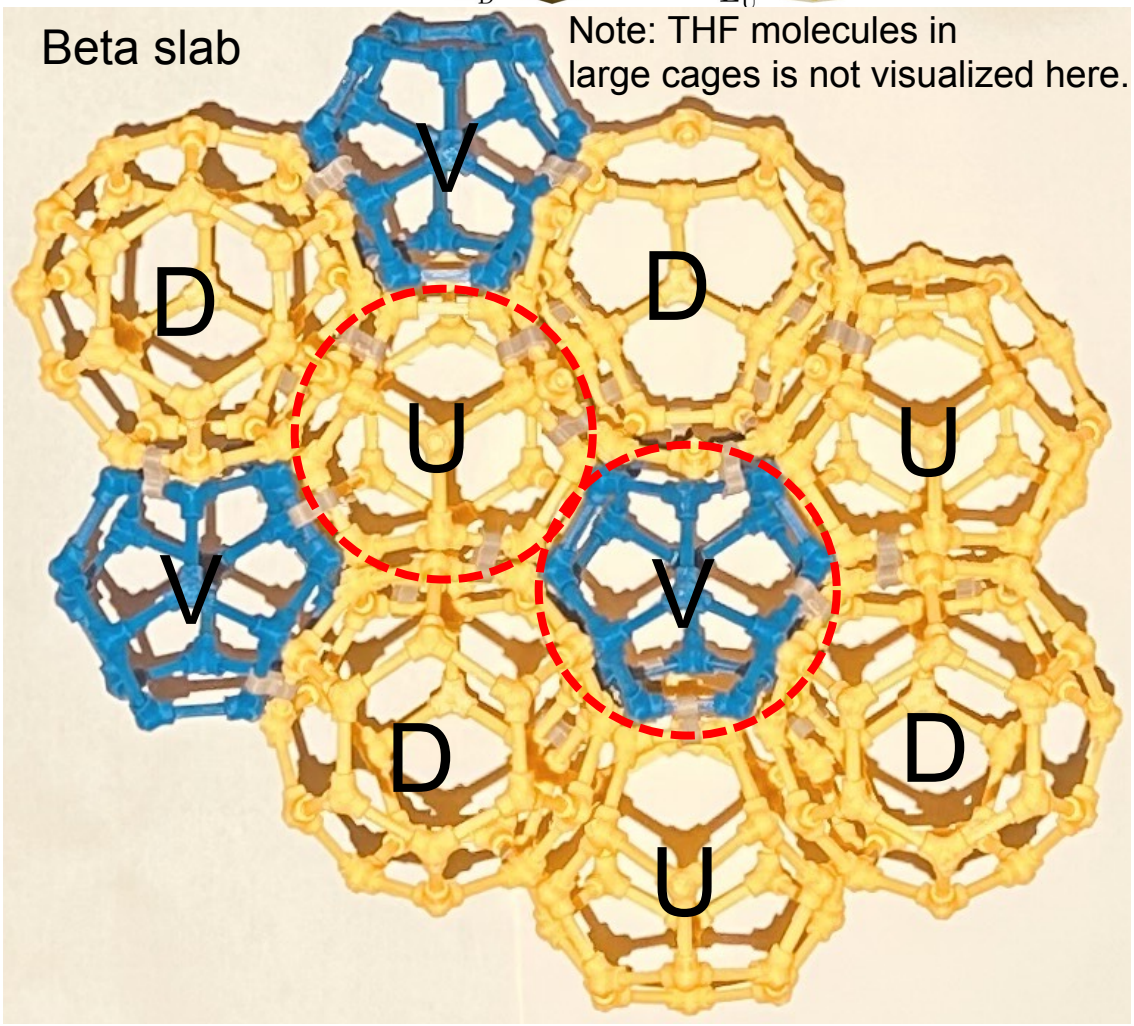
Hypothesis:
1. C corresponds to free OH in a water molecule.
2. T features correspond to THF molecules.
3. Q features also correspond to THF molecules at lower positions.

Considering symmetry we have **two options**:

Option 1: Centered at U
-Surrounded by both large and small cages, 3D and 3V cages.

Option 2: Centered at V
- Surrounded by large cages, 3 D and 3 U cages.

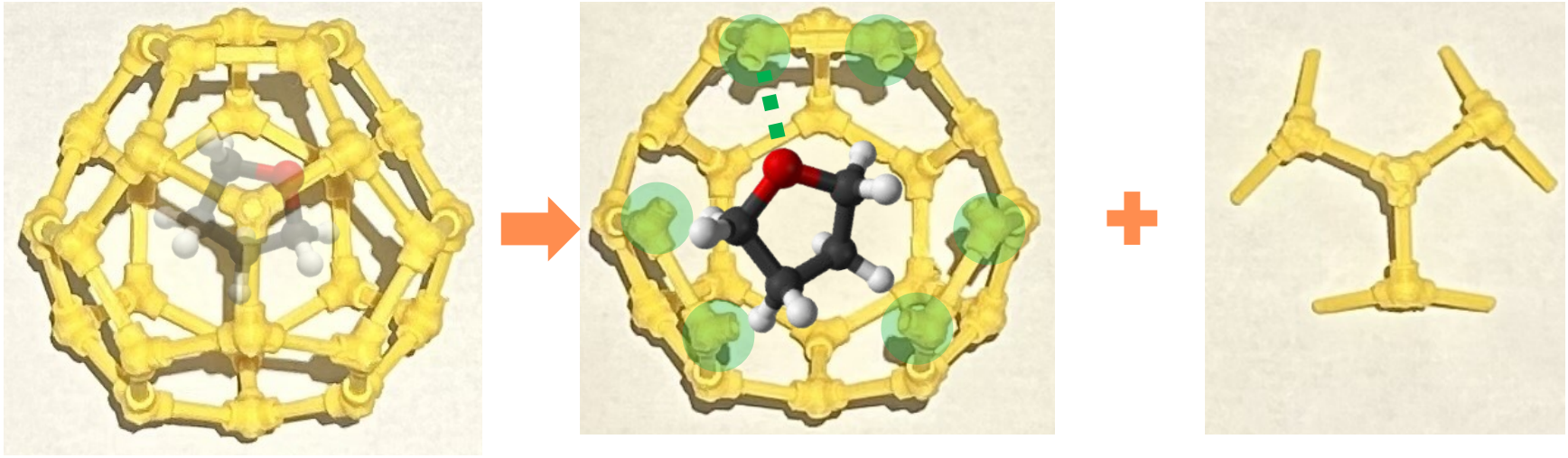




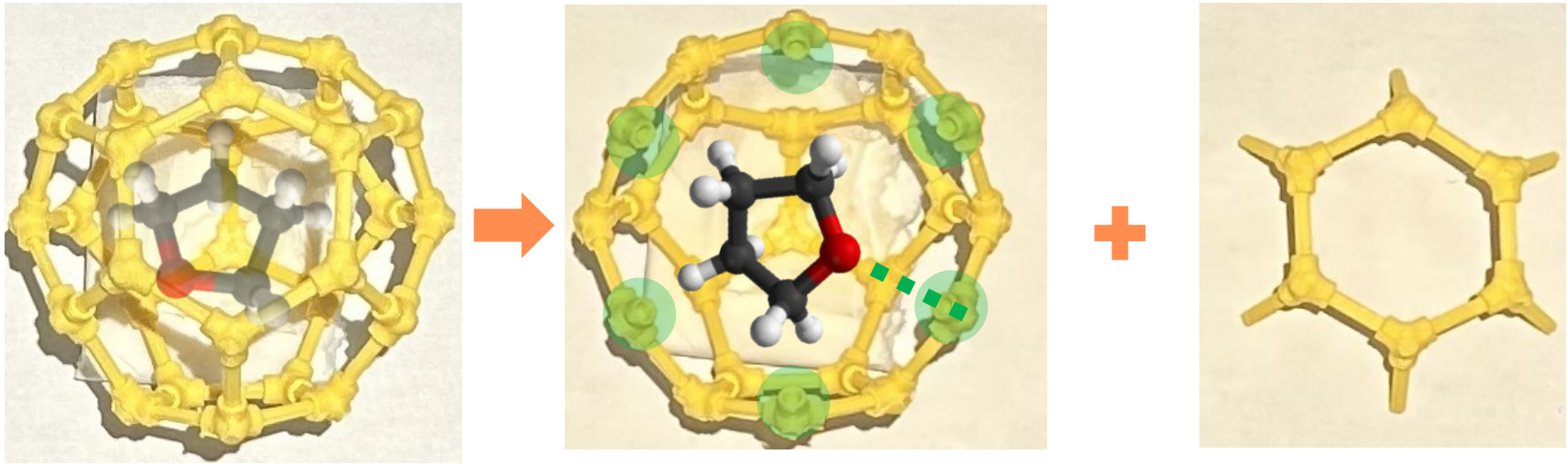
We chose option 2. To show the THF feature, **U and D needs to be disclosed**.

Removing top lid of large cage U and D

U



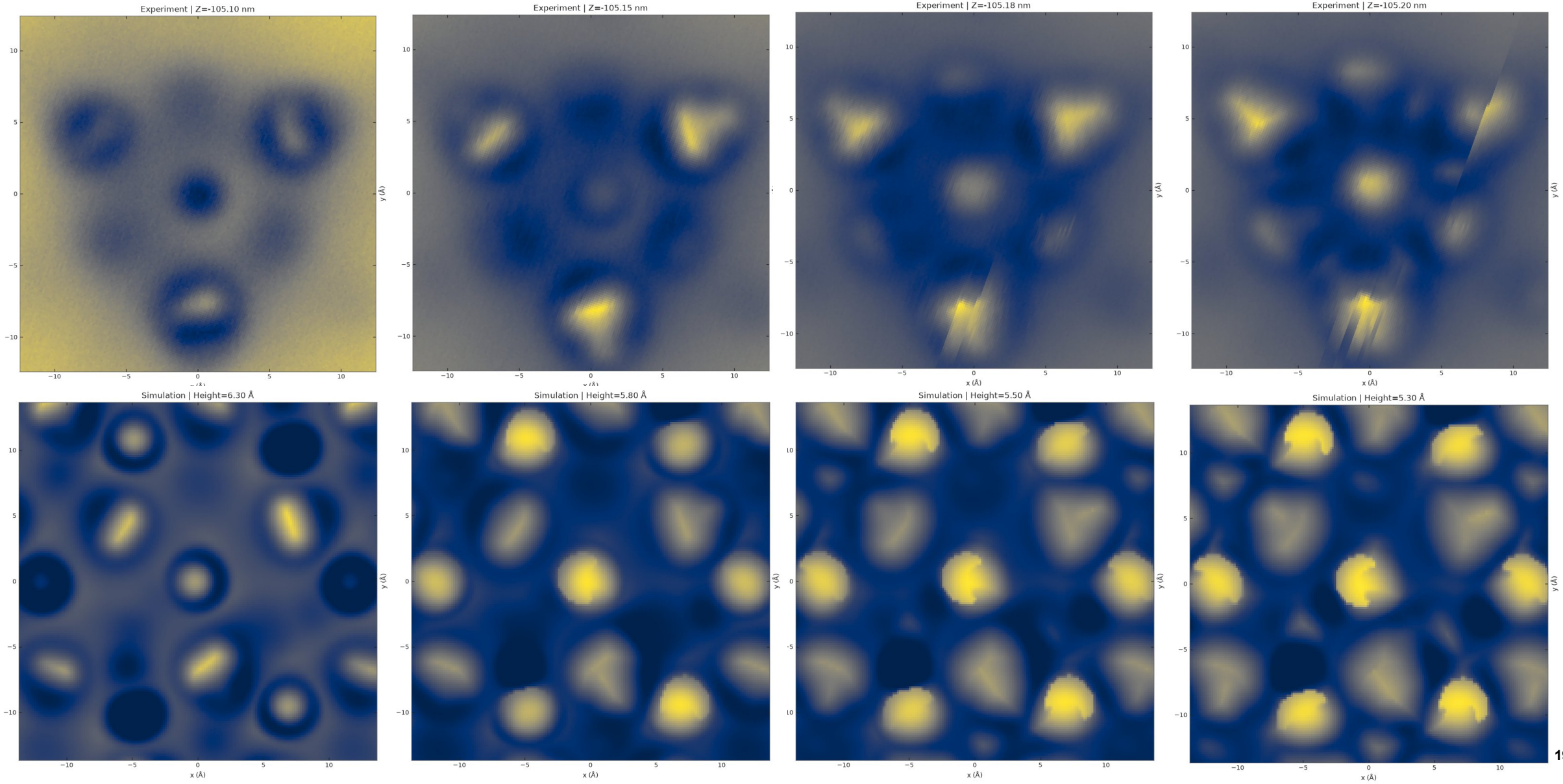
D



More molecules with free OH appears, the THF inside this cages are more likely to be lifted up, and H-bonded with extra OH.

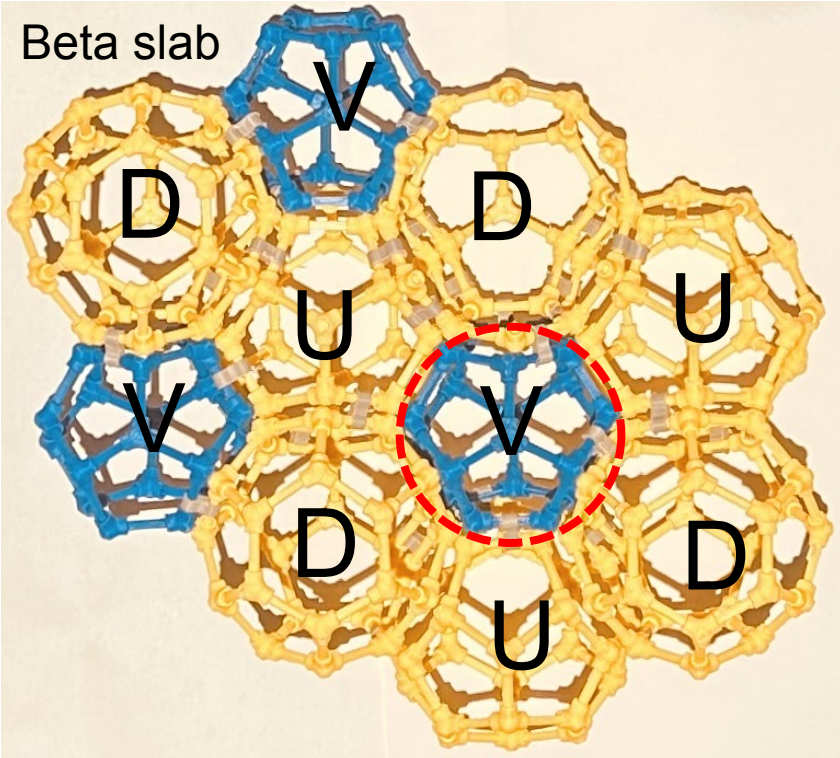
A!

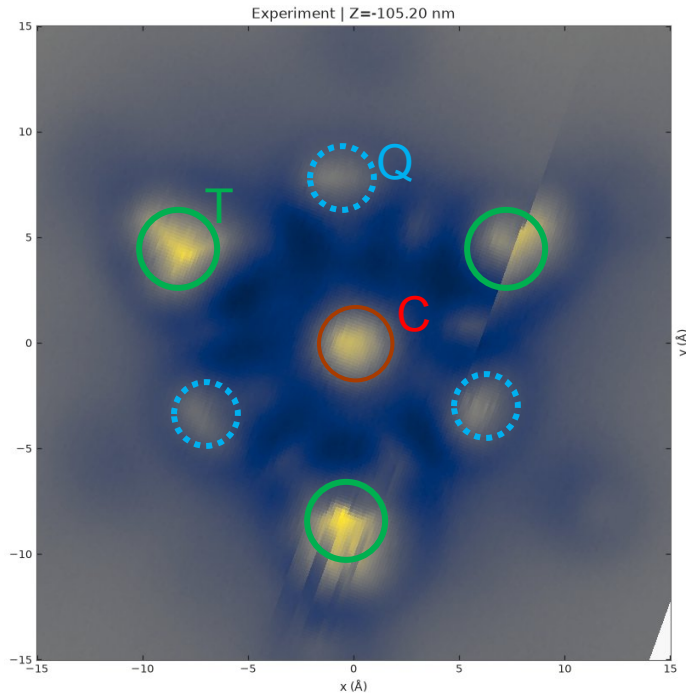
Simulations images compared to experimental images

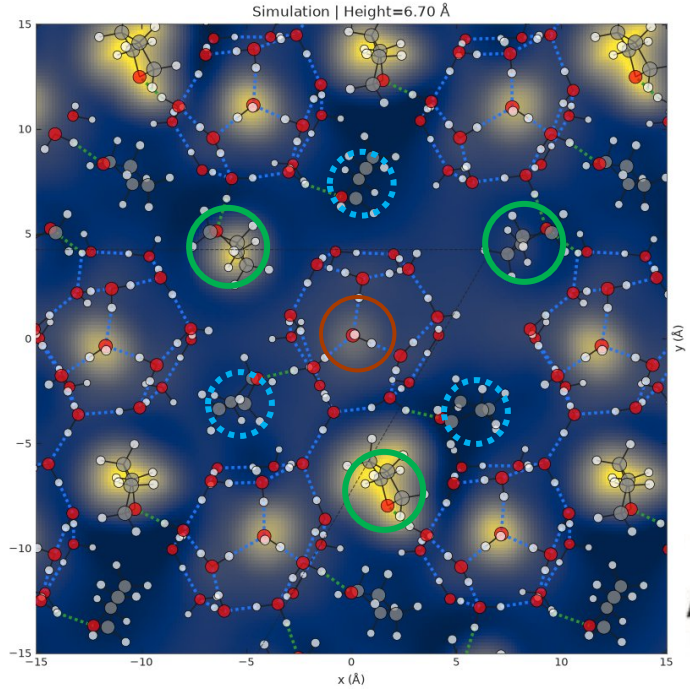


Simulation structure optimized by classical force fields; AFM simulated by PPM with point charges.

What we learn from the cluster structure?









Center feature C: the water molecules with free OH bond at the top vertex of small **V-orientated cage.**

T shape feature: the THF molecule with vertical orientation (θ is close to 90°).

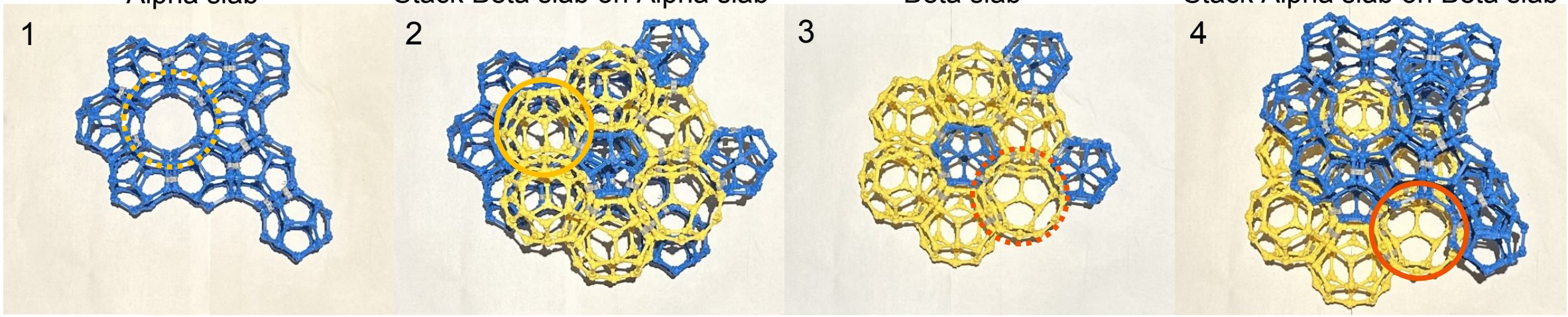
H-bonds are found between the THF and water molecules. The bonding sites affect (1) the distance d_{TC} to the centre feature.(2) The lateral orientation of T feature.

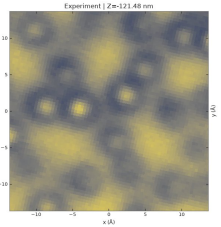
The experimental cluster corresponds to a fragment of Beta slab with large open cages.

A!

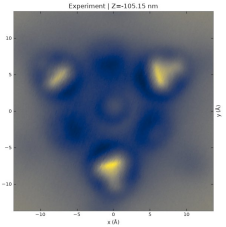
Summary

Alpha slab | Stack Beta slab on Alpha slab | Beta slab | Stack Alpha slab on Beta slab





The experimental surface is a complete Alpha slab with additional THF in open large cages.



The experimental cluster corresponds to a fragment of Beta slab with open large cages.

Extra THF in these open cages is needed in step 2, so it's very reasonable that extra THF is found in surface structure.
The experimental surface is post structure of Alpha slab.

To keep growing, all these open large cages need to be filled with THF and be closed.
The cluster structure is the pre-structure of Beta slab.

Proposed mechanism: surface growth is the repetition of steps 2 and 4 at a coarse-grained level.

A!

Acknowledgements

1. Advisor: Prof. **Adam S. Foster** (Aalto University, Finland).
2. Collaborators: **Yuejian Zhang**, Dr. **Jiani Hong**, and Prof. **Ying Jiang** (Peking University, China).
3. PPM developers: https://github.com/Probe-Particle/ppafm
4. World Premier International Research Center Initiative (WPI), MEXT, Japan.
5. Academy of Finland projects 347319 and 346824.
6. Computational resources: Aalto Science-IT Project and CSC.



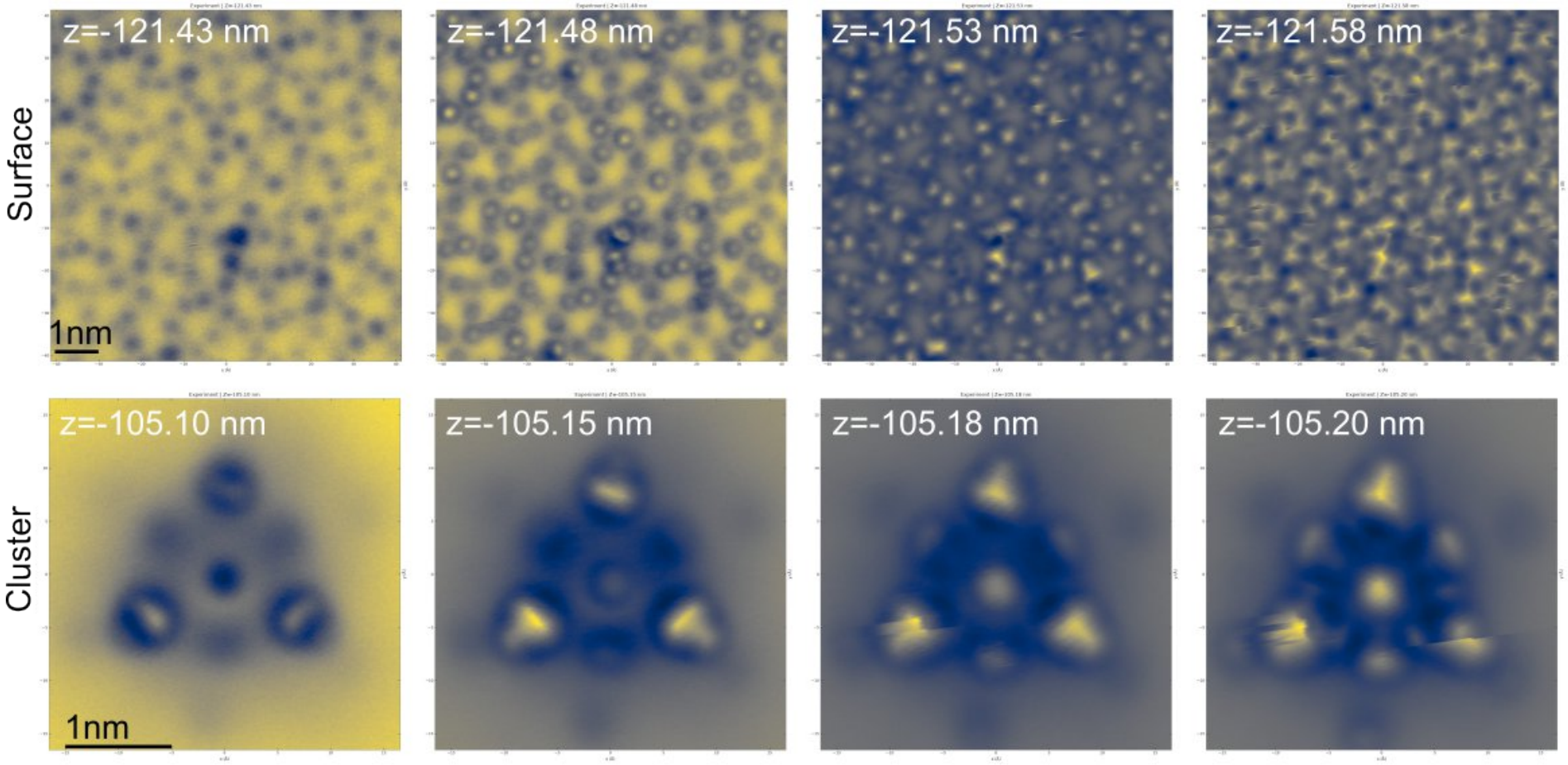
A!



Kiitos
aalto.fi

Appendix

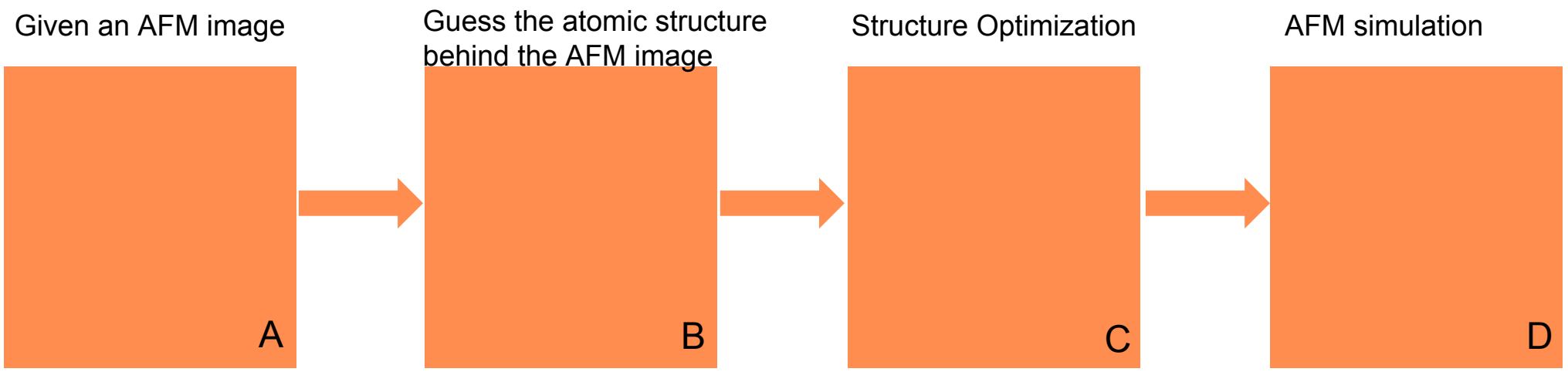
High-Resolution Height-Dependent AFM Images



The AFM images at different tip heights for the **two key stages: surface and cluster**.

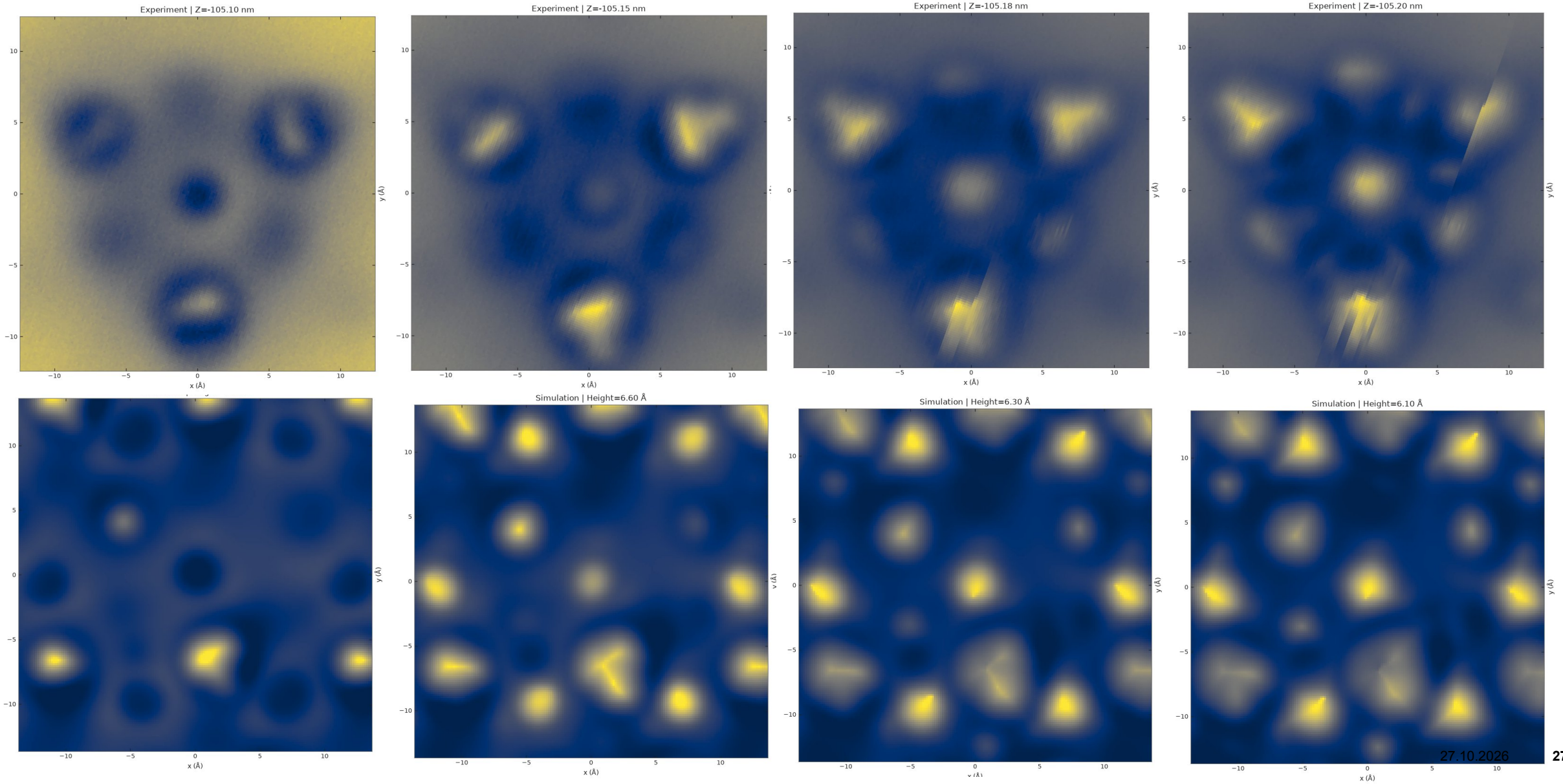
A!

Workflow to Find The Structure For a Given AFM Image



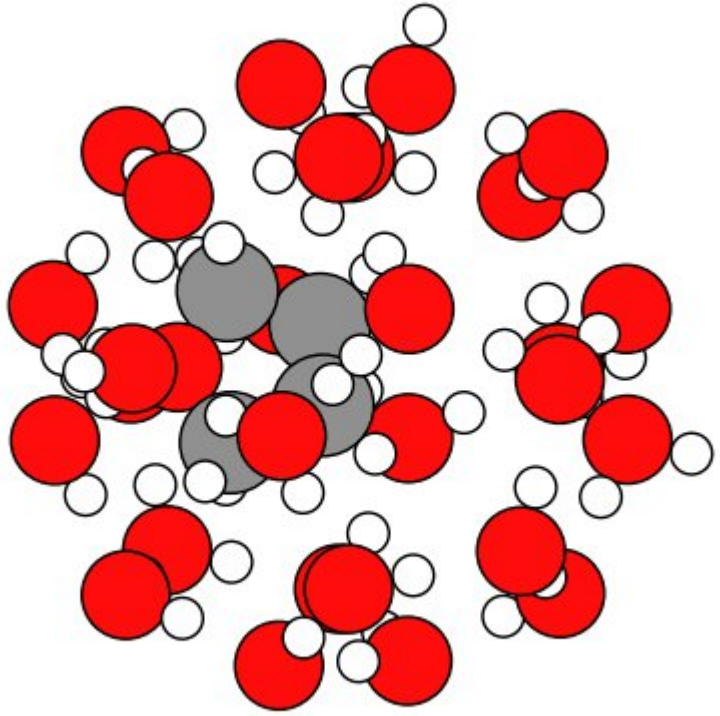
Repeat the processes of **B, C, and D**, until we find image **D and A are identical**, then we think **the structure of A is structure C**.

Simulations images compared to experimental images



Simulation structure re-optimized by DFT; AFM simulated by full density based PPM.

H-bonding in water and THF molecules



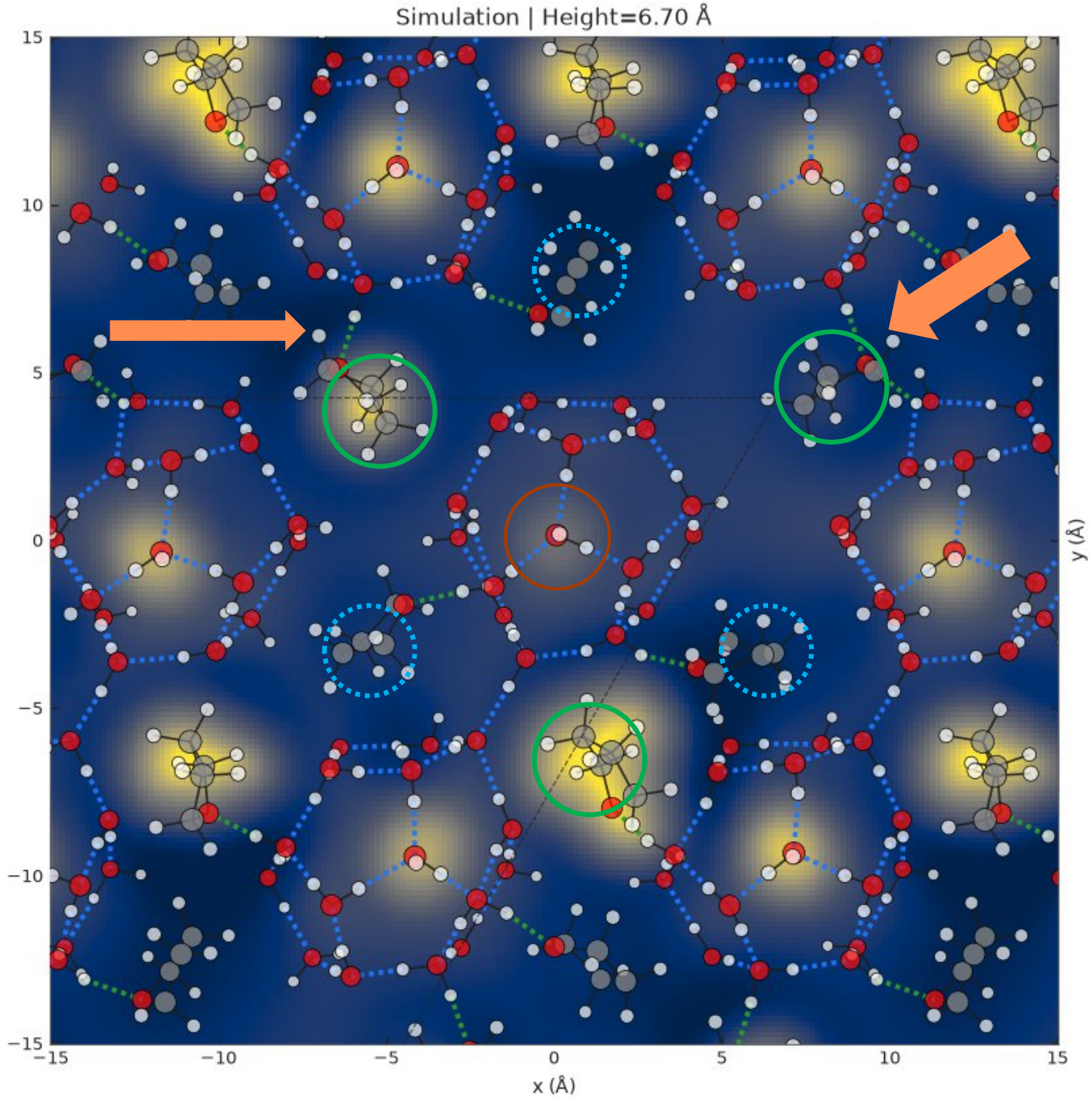
In the **closed large cages**, THF preferentially resides near the cage center. No hydrogen bonds are observed between THF and the surrounding water molecules, suggesting the **hydrophobic nature of the THF–water-cage interaction**.

However, this is not the same case when there are free OH when the cages is disclosed. **H-bond can be found between the THF and free OH in the disclosed cages.**.

A!

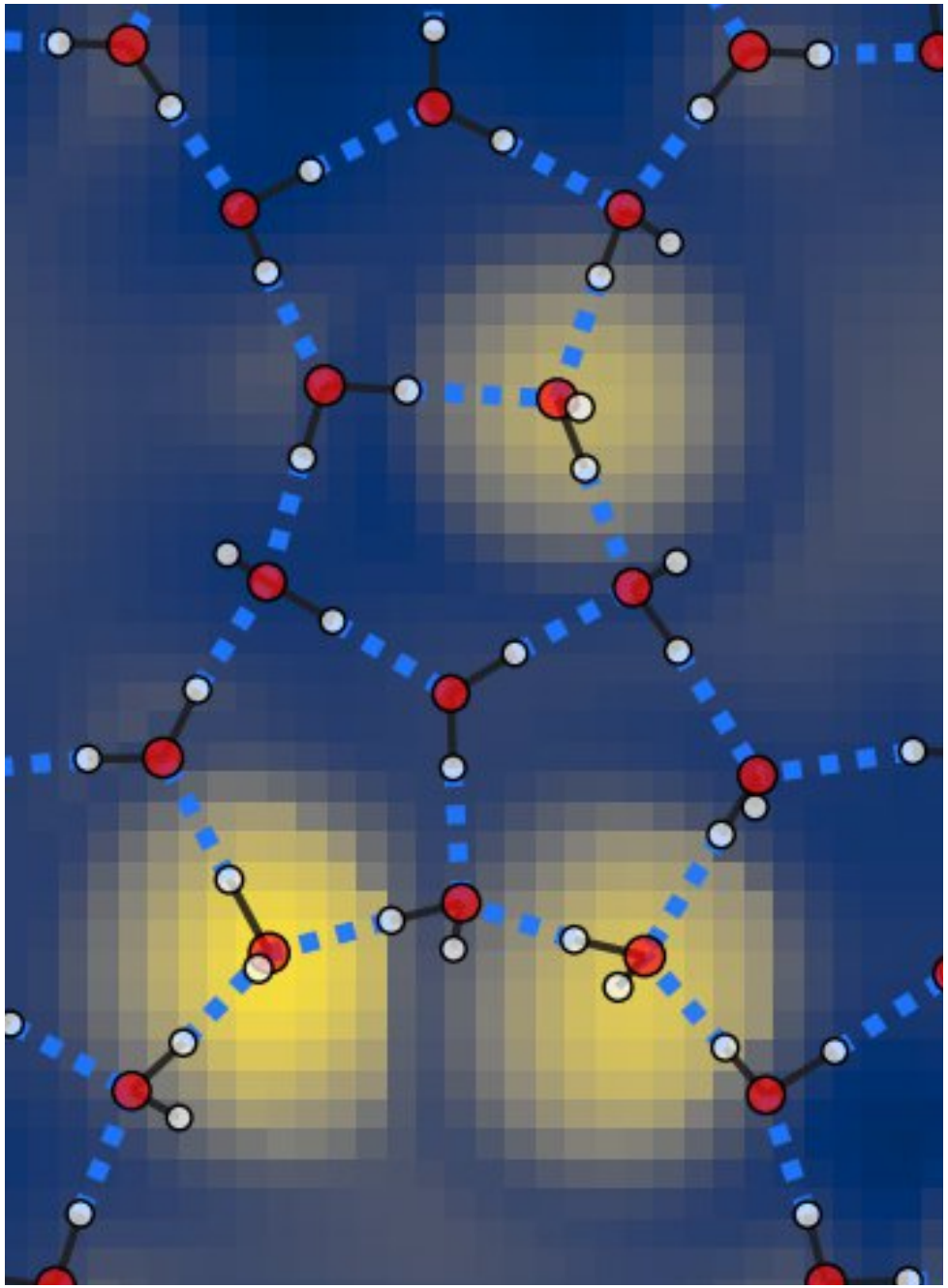
Note: The structure is in bulk system, other molecules besides this cage are not visualized.

H-bonds between THF (O) and Water (H-O)



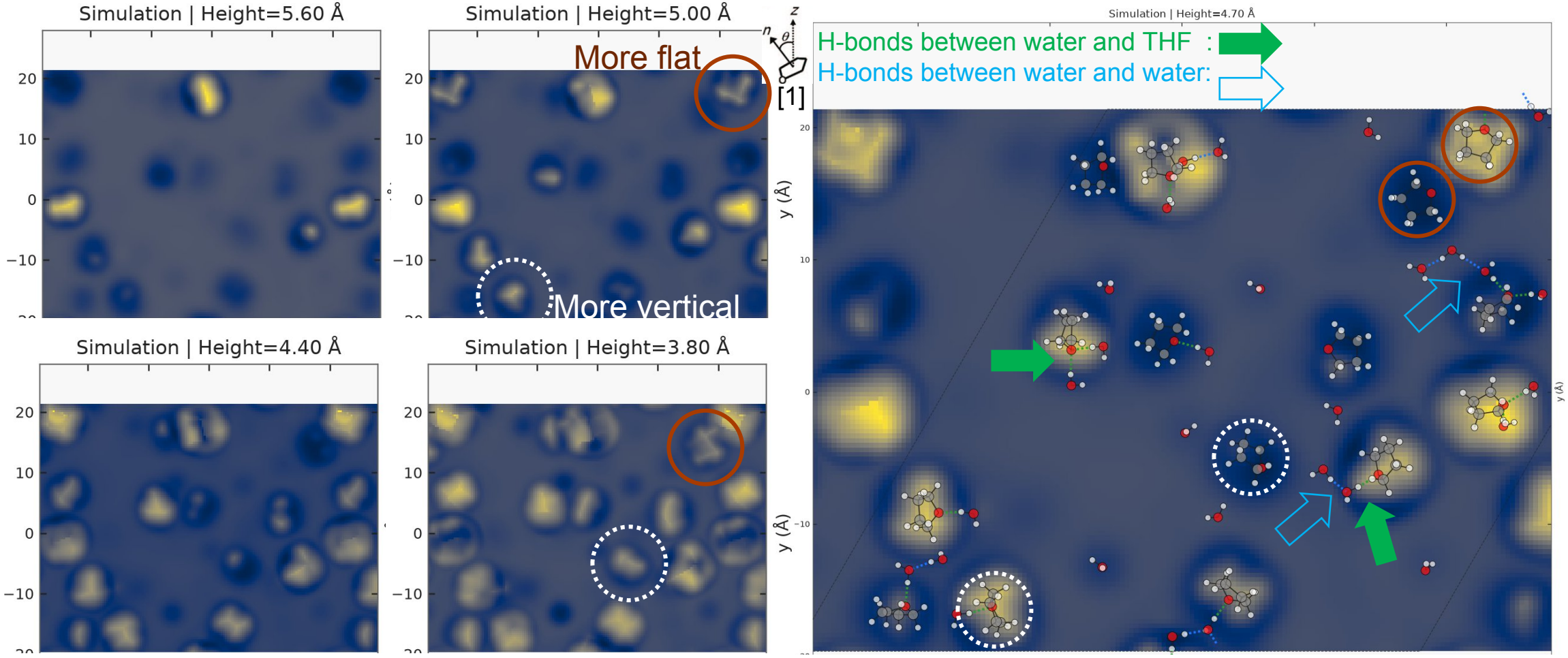
A!

Free OH



A!

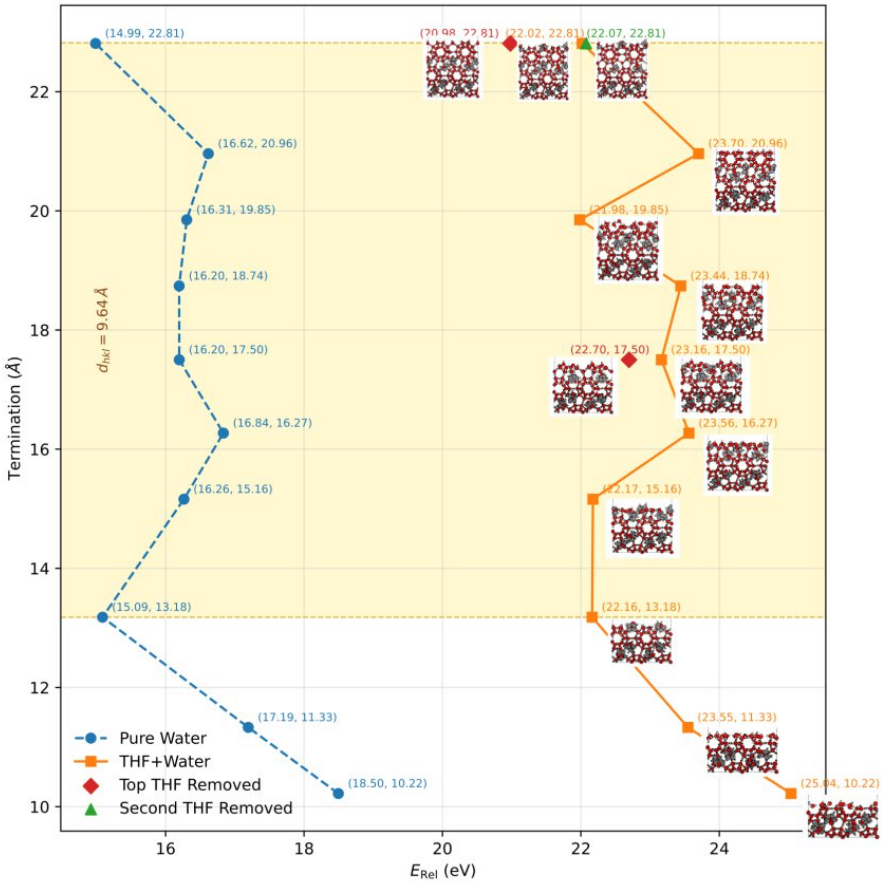
AFM simulations for THF molecules



H-bonds are found between the THF and water with free OH bond.
AFM **features of THF molecules are sensitive to its orientation**.

A!

Formation energies for different surface structure



Relative energy

$$E_{\mathrm{rel}} = E_{\mathrm{slab}} - \left(n_{\mathrm{H_2O}}\, E_{\mathrm{H_2O}} + n_{\mathrm{THF}}\, E_{\mathrm{THF}}\right)$$

where E_{slab} is the DFT energy of the relaxed slab, n_i the number of species i, and $E_{\mathrm{H_2O}}$, E_{THF} the isolated-molecule reference energies.

Termination	$n_{\mathrm{H_2O}}$	n_{THF}	E_{rel} (eV)
22.81 Å (top removed)	328	16	20.98
22.81 Å (2nd removed)	328	16	22.07
22.81 Å (with THF)	328	20	22.02
20.96 Å	304	16	23.70
19.85 Å	283	16	21.98
18.74 Å	272	16	23.44
17.50 Å	248	16	23.16
16.27 Å	238	16	23.56
15.16 Å	216	16	22.17

DFT simulations

- Structures were relaxed with periodic DFT (VASP 6.3.2) using the PAW method with PAW-PBE potentials for H, C, and O.
- The optB86b-vdW functional was used to describe dispersion interactions.
- Plane-wave cutoff: 500 eV. Large slab cells (up to $\sim$ 330 H_2O + $\sim$ 20 THF) were sampled at the Γ-point only.
- A Γ-centred $5 \times 5 \times 1$ mesh was used for the bulk reference cells.
- Geometries were optimised with the FIRE algorithm until residual forces < 0.01 eV/Å, with fixed lattice vectors.
- The relaxed structures were used to determine the bulk sII lattice parameter, build (111) surface slabs and island-like cluster models, and simulate high-resolution AFM images with the probe-particle model (ppafm) for direct comparison with experiment.

Classical force field simulations

- Because the systems are too large for DFT (up to $\sim$3000 atoms), structures were also explored with classical molecular dynamics (MD) using GROMACS.
- Water described with the **TIP4P-Ew** model [1]; other molecules (THF) with **OPLS-AA** parameters. The THF topology and charge distribution were generated with the LigParGen server (Jorgensen Lab) [2].
- Lattice-parameter scan of the bulk sII cell (a = 16.40–17.10 Å, Δa = 0.01 Å) $\rightarrow$ fitted optimum 16.75 Å, in agreement with DFT (16.70 Å).
- (111) surface (d_{111} = 9.67 Å) built by repeating the bulk cell $\times$3 along **c**; terminations cut by the O-density profile. Island-like cluster models ($\sim$3000 atoms) from expanded cells with surrounding molecules removed.
- Same analysis as DFT: bulk lattice parameter, terminations and cluster structures $\rightarrow$ AFM images via the probe-particle model (ppafm) with point charges, compared with experiment.

[1] J. Chem. Phys. 120, 9665 (2004)
[2] https://zarbi.chem.yale.edu/ligpargen/

AFM simulations

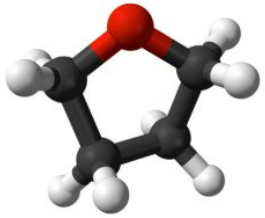
- High-resolution AFM images for a CO-terminated tip were simulated with the **full density-based model** (FDBM), as implemented in the probe-particle model (PPM, ppafm) with GPU acceleration.
- Inputs: atomic structure, electrostatic (Hartree) potential, and electron density, all calculated from DFT with VASP.
- Tip–sample interaction has three contributions:
  - **(i) Short range:** density-overlap model $V_{\mathrm{SR}} = V_0 \int [\rho^{\mathrm{tip}}(\vec{r})\, \rho^{\mathrm{sample}}(\vec{r})]^{\alpha}\, d\vec{r}$, with $V_0 = 18.0$ and $\alpha = 1.0$ (A_{Pauli}, B_{Pauli} in PPM);
  - **(ii) Electrostatic:** Hartree potential and tip charge density;
  - **(iii) van der Waals:** DFT-D3 scheme with PBE parametrisation.
- For large CFF-relaxed structures (∼3000 atoms), a simplified PPM with only vdW + point charges enables efficient comparison with experiment.

[1] FDBM: ACS Nano 13, 786 (2019); PPM: Comput. Phys. Commun. 305, 109341 (2024); Phys. Rev. B 90, 085421 (2014); Phys. Rev. Lett. 113, 226101 (2014).

AFM experimental setup

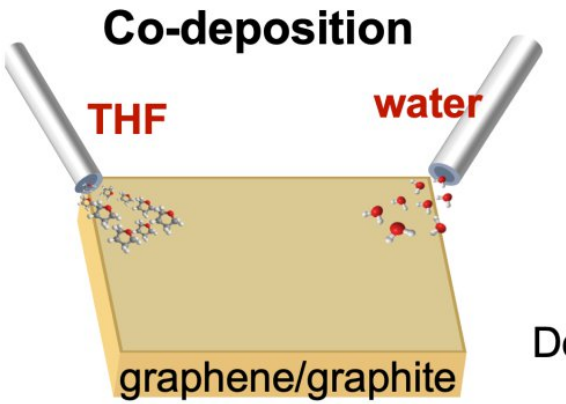
Ideal system: tetrahydrofuran(THF) hydrate

THF molecule



Conditions for hydrate formation

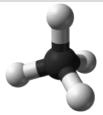
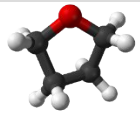
- Form around 4.4 ° with THF:H_2O=1:17
- Do not require high pressures
- Binary guest hydrates with gases



Deposition temperature: 110K

[1] Yuejian Zhang, Jiani Hong, Prof. Ying Jiang's group, Peking University (2025).

The formation conditions

Property	Methane Hydrate	THF Hydrate
Guest molecule	CH4 	THF 
Hydrate structure	Structure I	Structure II
Water/guest ratio	$CH_4 \cdot 5.75H_2O \approx CH_4 \cdot 6H_2O$	$THF \cdot 17H_2O$
Temperature	Usually below 10°C	Below about 4.4°C
Pressure	High pressure required	Forms at atmospheric pressure
Example condition	0°C, about 2.6 MPa	4.4°C, 0.1 MPa
Formation difficulty	Relatively difficult	Relatively easy

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